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BUS2 195B - Data Visualization/Exploration

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Project Task 3: Idea Development & ER Diagram

For this project, our group decided to center our idea around grocery sales trends. Specifically, we want to focus on analyzing various product categories, sales volume, and pricings over time in order to combat inefficient inventory management and shelf planning. By doing so, we hope to help grocery store owners and their managers identify the most popular products and seasonal trends, with the intention of improving efficiency and effectiveness. Ultimately, this project would allow us to identify the different factors that are affecting sales performance and how to help store managers plan more effectively. Overall, we aim at providing our knowledge and resources to assist grocery store stakeholders and the managers that help run their locations.

One product belongs to one category
One category have many products

One product can have 0 or many sales
One sale belong to one product

One employee can do many sales
Sale can only be done by one employee

One city can have none or many employees
One employee belong to one city

products	
PK	<u>ProductID</u>
	ProductName
	Price
	CategoryID
	Class
	ModifyDate
	Resistant
	IsAllergic
	VitalityDays

sales	
PK	<u>SalesID</u>
FK	SalesPersonID
FK	CustomerID
FK	ProductID
	Quantity
	Discount
	TotalPrice
	SalesDate
	TransactionNumber

employees	
PK	<u>EmployeeID</u>
	FirstName
	MiddleInitial
	LastName
	BirthDate
	Gender
FK	CityID
	HireDate

categories	
PK	<u>CategoryID</u>
	CategoryName

customers	
PK	<u>CustomerID</u>
	FirstName
	MiddleInitial
	LastName
FK	CityID
	Address

cities	
PK	<u>CityID</u>
	CityName
	Zipcode
FK	CountryID

countries	
PK	<u>CountryID</u>
	CountryName
	CountryCode

One city can have none or many customers
One customer belong to one city

One country have many cities
One city belong one countries

One customer can have many sales
One sale can only be one customer